

ELI5 Why is water conservation and ai data centers using water so important if we can just reuse that water easily?

Posted by u/passisgullible • 0 points • 86 comments

Like water is infinite since it's recycled why is not wasting it so important? TIA!

Comments

u/explainlikeimfive-ModTeam • 1 points • 2026-01-28 01:16 UTC

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u/_Aj_ • 1 points • 2026-01-28 01:14 UTC

This was asked 3 years ago and an expert replied:

> https://www.reddit.com/r/explainlikeimfive/comments/yzlsoj/eli5_why_do_datacenters_continuously_use_more/

Basically evaporative cooling. Which by nature loses water. If that's coming from rivers or reservoirs it's additional strain on that system. A data centre may use as much water as 6000 homes would use. There's also the cost of water for the chips manufactured to go the servers installed there and the power it uses which are indirect water uses.

<https://www.eesi.org/articles/view/data-centers-and-water-consumption#:~:text=A%20federal%20report%20estimated%20that,of%20ultrapure%20water%20per%20day.>

[https://theconversation.com/data-centers-consume-massive-amounts-of-water-companies-rarely-tell-the-public-exactly-how-much-262901#:~:text=They%20use%20water%20directly%20for%20cooling%2C%20and,\(2.9%20billion%20liters\)%20used%20by%20data%20centers.](https://theconversation.com/data-centers-consume-massive-amounts-of-water-companies-rarely-tell-the-public-exactly-how-much-262901#:~:text=They%20use%20water%20directly%20for%20cooling%2C%20and,(2.9%20billion%20liters)%20used%20by%20data%20centers.)

> Google reports that in 2024, the company's data center in Council Bluffs, Iowa, consumed 1 billion gallons of water (3.8 billion liters), the most of any of its data centers.

The average home uses 100,000 gal a year. So that's approx ****10 million homes**** worth of water use per year from Google's largest data center.

u/wimpies • 1 points • 2026-01-28 01:13 UTC

Honest answer is that it isn't.

Data Centers use a mix 3 water sources typically.

- 1) They have their own wells
- 2) They use non-potable (non-drinking) water
- 3) They use municipal, clean water.

And above that there is generally two ways water is used

- A) In a closed loop cooling system
- B) In evaporative cooling towers.

Out of those 6 possible combinations only "3B" is the one that "uses up" water that could have been better used elsewhere. And even then, a lot of the places where Data Centres are built so t have water scarcity issues and/or if they do they avoid that option.

That's not to say it isn't a problem than can't be fixed. It definitely can be. But the issue of "water use" in data centres is vastly overstated often by people who are not knowledgeable in the subject and/or have an agenda to push.

u/Writeous4 • 1 points • 2026-01-28 01:11 UTC

The answers being given here aren't wrong (mostly) in the abstract - the water on Earth is generally fixed and reusable but it does take resources to access and recycle, etc.

However, an important caveat is - water usage really *isn't* an issue with AI nor data centres. It's mostly an invented issue. There are other concerns e.g their energy usage, but water used for AI is really not that significant and has been subject to intense misinformation.

u/sessamekesh • 2 points • 2026-01-28 01:09 UTC

Water that falls from the sky is pretty much free. Water that comes out of a pipe isn't. AI data centers use the second one.

u/tortuga8831 • 2 points • 2026-01-28 01:08 UTC

https://youtu.be/H_c6MWk7PQc?si=Lq7LgnT-cbY-_rxz Hank green touches on this topic and gives a surface explanation.

u/explainlikeimfive-ModTeam • 1 points • 2026-01-28 01:17 UTC

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u/Lazy_Permission_654 • 0 points • 2026-01-28 01:08 UTC

It's a dishonest talking point because not all datacenters consume water, not all consume potable water and not all uses of AI consume meaningful amounts

It's a complex issue that cannot be meaningfully summarized and all locations, systems and infrastructure are built differently

The solution to the consumption is to reward those who are neutral and disincentive consumption

u/Diligent_Explorer717 • 1 points • 2026-01-28 01:07 UTC

Many times it's about creating a process of accountability, if we don't start 'policing' tech bros and ai now, even over 'small' things, there's no limit to what they would do to the environment to further their goals.